Q4 冷媒性能比较 / Refrigerant comparison

【1】逐图计算 / Calculations from the P-H diagrams

Refrigeration effect (q_e) = $h_1 - h_3$ (kJ/kg)

$$Coefficient\ \ of\ \ Performance\ \ (COP) = \frac{h_1 - h_3}{h_2 - h_1}$$

图 1 / Figure 1 — R-22

量及计算 / Quantity and calculation	Chiller: 5℃ / 45℃ 蓝色 / Blue	Room Cooler: 10℃ / 60℃ 橙色 / Orange
h_1 / kJ/kg	408	408.5
h_2 / kJ/kg	435	438
h_3 / kJ/kg	256	278
Refrigeration effect (q_e) h_1–h_3 / kJ/kg	408 − 256 = 152	408.5 − 278 = 130.5
Compression work (w) h_2–h_1 / kJ/kg	435 − 408 = 27	438 − 408.5 = 29.5
Coefficient of Performance (COP)	$\frac{408 - 256}{435 - 408} = 5.63$	$\frac{408.5 - 278}{438 - 408.5} = 4.42$
Evaporating pressure p_e / bar(abs)	6	7
Condensing pressure p_c / bar(abs)	17	24.1
Specific volume (inlet) v_1 / m^3/kg	0.040	0.034
Specific volume (outlet) v_2 / m^3/kg	0.0155	0.0108
Entropy s / kJ/(kg·K)	1.75	1.73

图 2 / Figure 2 — R-407C

量及计算 / Quantity and calculation	Chiller: 5℃ / 45℃ 橙色 / Orange	Room Cooler: 10℃ / 60℃ 绿色 / Green
h_1 / kJ/kg	415	417
h_2 / kJ/kg	446	449
h_3 / kJ/kg	272	304
Refrigeration effect (q_e) h_1–h_3 / kJ/kg	415 − 272 = 143	417 − 304 = 113
Compression work (w) h_2–h_1 / kJ/kg	446 − 415 = 31	449 − 417 = 32
Coefficient of Performance (COP)	$\frac{415 - 272}{446 - 415} = 4.61$	$\frac{417 - 304}{449 - 417} = 3.53$
Evaporating pressure p_e / MPa(abs)	0.55	0.64
Condensing pressure p_c / MPa(abs)	1.9	2.8
Specific volume (inlet) v_1 / m^3/kg	0.043	0.038
Specific volume (outlet) v_2 / m^3/kg	0.013	0.0082
Entropy s / kJ/(kg·K)	1.77	1.76

图 3 / Figure 3 — R-134a

量及计算 / Quantity and calculation	Chiller: 5℃ / 45℃ 绿色 / Green	Room Cooler: 10℃ / 60℃ 橙色 / Orange
h_1 / kJ/kg	401	404
h_2 / kJ/kg	426	434
h_3 / kJ/kg	263	287
Refrigeration effect (q_e) h_1–h_3 / kJ/kg	401 − 263 = 138	404 − 287 = 117
Compression work (w) h_2–h_1 / kJ/kg	426 − 401 = 25	434 − 404 = 30
Coefficient of Performance (COP)	$\frac{401 - 263}{426 - 401} = 5.52$	$\frac{404 - 287}{434 - 404} = 3.90$
Evaporating pressure p_e / MPa(abs)	0.34	0.41
Condensing pressure p_c / MPa(abs)	1.2	1.7
Specific volume (inlet) v_1 / m^3/kg	0.060	0.053
Specific volume (outlet) v_2 / m^3/kg	0.017	0.014
Entropy s / kJ/(kg·K)	1.73	1.72

图 4 / Figure 4 — R-123

量及计算 / Quantity and calculation	Chiller: 5℃ / 45℃ 蓝色 / Blue	Room Cooler: 10℃ / 60℃ 橙色 / Orange
h_1 / kJ/kg	384	386
h_2 / kJ/kg	407	414
h_3 / kJ/kg	245	261
Refrigeration effect (q_e) h_1–h_3 / kJ/kg	384 − 245 = 139	386 − 261 = 125
Compression work (w) h_2–h_1 / kJ/kg	407 − 384 = 23	414 − 386 = 28
Coefficient of Performance (COP)	$\frac{384 - 245}{407 - 384} = 6.04$	$\frac{386 - 261}{414 - 386} = 4.46$
Evaporating pressure p_e / MPa(abs)	0.04	0.051
Condensing pressure p_c / MPa(abs)	0.18	0.28
Specific volume (inlet) v_1 / m^3/kg	0.38	0.30
Specific volume (outlet) v_2 / m^3/kg	0.097	0.058
Entropy s / kJ/(kg·K)	1.67	1.66

【2】逐项比较及构造影响 / Comparison and construction implications

（1）Coefficient of Performance (COP)｜制冷性能系数

按图中数据，两工况均为 R-123＞R-22＞R-134a＞R-407C；房间工况下 R-123 与 R-22 很接近。同制冷量下，COP 越高，压缩机耗功及冷凝器散热量越小。由 5/45 改为 10/60 工况，温升由 40K 增至 50K，各 COP 均下降。

For these diagram readings, both cases rank R-123 > R-22 > R-134a > R-407C; R-123 and R-22 are close in the room-cooler case. Higher COP reduces compressor power and condenser heat rejection at equal duty. The 40K-to-50K increase in temperature lift lowers all four COPs.

（2）Refrigeration effect｜单位质量制冷效应

R-22 两工况均最大，分别为 152 及 130.5kJ/kg。同制冷量下，较大的 q_e 使所需质量流量 $\dot{m}=\dot{Q}_e/q_e$ 较小。房间工况中，四种冷媒的 q_e 均降低、单位质量压缩功 w 均增加。

R-22 has the greatest effect in both cases: 152 and 130.5kJ/kg. At equal cooling duty, larger q_e reduces mass flow, $\dot{m} = \dot{Q}_e/q_e$. In the room-cooler case, all four have lower q_e and greater specific compressor work w.

（3）Evaporating pressure｜蒸发压力

两工况均为 R-22＞R-407C＞R-134a＞R-123。R-123 仅 0.040 及 0.051MPa(abs)，低于大气压，容易吸入空气及水分，故须良好密封、排不凝气体装置及水分控制；排气不能代替修漏。

Both cases rank R-22 > R-407C > R-134a > R-123. R-123 evaporates at only 0.040 and 0.051MPa(abs), below atmospheric pressure. It requires good sealing, a purge unit and moisture control to limit inward air/moisture leakage; purging does not replace leak repair.

（4）Condensing pressure｜冷凝压力

两工况均为 R-407C＞R-22＞R-134a＞R-123。R-407C 最高，为 1.9 及 2.8MPa(abs)（19 及 28bar），其次 R-22 为 17 及 24.1bar(abs)；其管道、阀件和换热器须有相应耐压等级。R-123 的低运行压力不免除超压保护。

Both cases rank R-407C > R-22 > R-134a > R-123. R-407C is highest at 1.9 and 2.8MPa(abs) (19 and 28bar), followed by R-22 at 17 and 24.1bar(abs). Their pipes, valves and heat exchangers need corresponding pressure ratings. Low operating pressure does not remove R-123's need for overpressure protection.

（5）Specific volume｜压缩机入口及出口比容

各组均 $v_2＜v_1$。R-123 的压缩机入口及出口比容远大于其余三种；同冷量下应按 $\dot{V}_1=\dot{Q}_e v_1/q_e$ 确定吸气容积流量。其大容积流量、低压力特点有利于采用 Centrifugal compressor；R-22/R-407C 较利于紧凑型机组，R-134a 居间。

Every case has $v_2 < v_1$. R-123 has much larger specific volumes at compressor inlet and outlet. At equal duty, suction volume flow is $\dot{V}_1 = \dot{Q}_e v_1/q_e$. Its high volume, low pressure application favours a Centrifugal compressor. R-22/R-407C favour compact units, with R-134a intermediate.

（6）Entropy｜熵

各图按 $s_1=s_2$ 确定理想等熵压缩终点。实际绝热压缩会产熵，通常需要更多功，COP 较低；不同冷媒的绝对熵值不能直接作为效率排名。R-123 图示终点接近／略入湿区，实际须保证适当吸气过热以避免湿压缩。

Each plot uses $s_1 = s_2$ to locate the ideal isentropic outlet. Real adiabatic compression generates entropy, requires more work and lowers COP. Absolute entropy values do not rank different refrigerants' efficiency. R-123's plotted outlets approach/slightly enter the wet region, so suitable suction superheat is needed to avoid wet compression.

工况及换热器 / Conditions and heat exchangers

R-407C 按图中饱和汽、液端点，将蒸发标温理解为点 1 露点、冷凝标温理解为点 3 泡点；水平段是等压段，相变沿程有温度滑移。

For R-407C, the plotted vapour/liquid endpoints are interpreted as the state-1 evaporating dew point and state-3 condensing bubble point. Horizontal processes are isobars; phase change has temperature glide.

【3】安全及环境影响 / Safety and environmental impacts

（7）Corrosion｜腐蚀性

四者均须配用相容的材料及润滑油，并保持系统干燥、洁净；水分、污染及高温分解物可引起腐蚀。R-123 低压侧较易吸入空气及水分，尤须密封及干燥控制。

All four require compatible materials/lubricant and a dry, clean system. Moisture, contamination and thermal decomposition products can cause corrosion. R-123's sub-atmospheric side especially needs effective sealing and moisture control.

（8）Explosion｜爆炸及爆裂风险

四者均为安全分类 1 级，即标准试验下无火焰传播，但仍可能因高温、超压造成设备爆裂。所有机组均须有适当耐压、压力切断及泄压保护，较高压力的 R-22/R-407C 尤其须关注。

All four are class 1, with no flame propagation in classification tests, but heat/overpressure can still rupture equipment. All systems need suitable pressure ratings, pressure cut-outs and relief protection, particularly the higher-pressure R-22/R-407C systems.

（9）Toxicity｜毒性

R-123 为 B1，毒性类别高于 R-22、R-134a 及 R-407C 的 A1。四者均须控制泄漏并通风；R-123 尤其须注意人员暴露和泄漏检测。高浓度泄漏还可能造成缺氧。

R-123 is B1, a higher toxicity class than A1 R-22, R-134a and R-407C. Control leaks and provide ventilation for all four, with particular attention to R-123 exposure and detection. High concentrations can also displace oxygen.

（10）Ozone Depletion Potential (ODP)｜臭氧层消耗潜能

R-22 及 R-123 为含氯 HCFC，课件 ODP 分别为 0.05 及 0.014；R-134a 及 R-407C 为 HFC，ODP 为 0。R-22/R-123 泄漏会破坏臭氧层，故须控制泄漏并回收冷媒。

Chlorine-containing HCFCs R-22 and R-123 have course-table ODPs of 0.05 and 0.014. HFCs R-134a and R-407C have zero ODP. R-22/R-123 leakage damages the ozone layer, requiring containment and refrigerant recovery.

（11）Global Warming Potential (GWP)｜全球变暖潜能

统一采用 IPCC AR4 的 100 年 GWP：R-22 为 1810、R-123 为 77、R-134a 为 1430、R-407C 约 1774。四者泄漏均造成直接温室效应，R-123 较低；较高 COP 可减少耗电引致的间接排放。

On the same IPCC AR4 100-year basis, GWP is 1810 for R-22, 77 for R-123, 1430 for R-134a and about 1774 for R-407C. All cause direct warming when leaked, with R-123 lower. Higher COP reduces electricity-related indirect emissions.

应用 / Application

Chiller 可采用 Shell and tube Water Chiller 结构：大型机组用 Flooded type（或 semi-flooded type），小型机组可用 Direct expansion type；Room Cooler 采用 Direct Expansion (DX) Coil。R-407C 温度滑移须与换热器及 Thermal Expansion Valve (TEV)控制匹配。

Chiller applications can use a Shell and tube Water Chiller: Flooded type (or semi-flooded type) for larger chillers and Direct expansion type for smaller chillers. Room Cooler applications use a Direct Expansion (DX) Coil. R-407C glide requires suitable heat exchange and Thermal Expansion Valve (TEV) control.

资料 / Sources: 手绘循环数据及 DuPont P–H 图；HVAC System II, slides 9–20, 29, 44–48；ASHRAE Handbook—Fundamentals (2025), Ch.29；IPCC AR4 (2007), WGI Table 2.14, pp.212–213（GWP100；R-407C 按 R32/R125/R-134a 的 23/25/52 质量分数加权 / mass-weighted）。